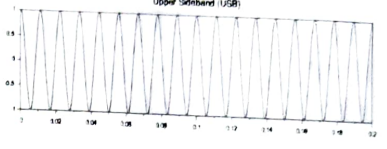
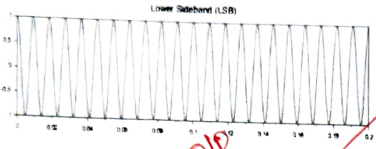
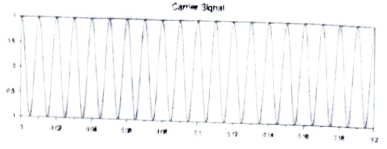
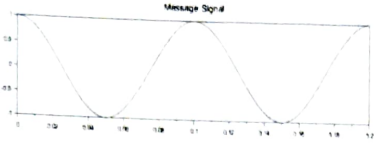


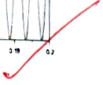
Graph Window Number 1

File Edit View

Graph Window Number 2



OK



27/7/26

exp3 SINGLE SIDED BAND SUPRESSED CARRIER:-

AIM:-

To simulate single sided band-suppressed carrier (SSB-SC) modulation and its Bandwidth.

EQUIPMENT REQUIRED:-

Scilab software

scilab program (20) * Implementat ⁿ	Waveforms and Tabulation (20)	calculation and result (20)	Viva- Voce (20)	Result/ observat ⁿ (20)

CODE:-

```
clc;
clear;
close;
```

```
Am=2;
```

```
Ac=4;
```

```
fm=2000;
```

```
fc=2000;
```

```
t = 0:0.00001 : 0.02;
```

```
em1 = Am * cos(2 * %pi * fm * t);
```

```
em2 = Am * cos(2 * %pi * fc * t + %pi/2);
```

```
ec1 = Ac * cos(2 * %pi * fc * t);
```

```
ec2 = Ac * cos(2 * %pi * fc * t + %pi/2);
```

S.No	Particular	Theory		Practical		
		Amplitude	Frequency	Amplitude	Time	frequency.
1.	Message	2	200	2	$T = x_2 - x_1$ $\Rightarrow 0.100 - 0.000$ $= 0.1001s$	$F = 1/T$ $= 10Hz$
2.	Carrier	45	4000 ²⁰⁰	4	$T = x_2 - x_1$ $\Rightarrow 0.010 - 0.000$ $= 0.0103s$	$F = 1/T$ $= 100Hz$
3.	USB	1 —	$f_c + f_m$ $= 110$	$f_c + f_m =$ $4400Hz$	$T = x_2 - x_1$ $\Rightarrow 0.00909 -$ 0.000 $= 0.00909s$	$F = 1/T$ $\Rightarrow 110Hz$
4.	LSB	1 —	$f_c - f_m$ $= 40$	$f_c - f_m =$ $3600Hz$	$T = x_2 - x_1$ $= 0.01111 -$ 0.000 $= 0.01111s$	$F = \frac{1}{T}$ $\Rightarrow 90Hz$

$$DSBSC1 = em1 * ec1;$$

$$DSBSC2 = Am2 * e22;$$

$$LSB = DSBSC1 + DSBSC2;$$

$$USB = DSBSC1 - DSBSC2;$$

subplot(3,2,1);

plot(t, em1);

title("Message signal");

subplot(3,2,2);

plot(1, em1);

title("Carrier signal");

subplot(3,2,3);

plot(t, LSB);

title("Lower sideband (LSB)");

subplot(3,2,4);

plot(t, USB);

title("Upper sideband (USB)");

(CALCULATION:-

$$Ma_{(theory)} = \frac{E_m}{E_c} \Rightarrow \frac{1}{1} \Rightarrow 1 \quad \frac{2}{4} \Rightarrow 0.5$$

$$Ma_{(practical)} \Rightarrow \frac{E_{max} - E_{min}}{E_{max} + E_{min}} = 0.5$$

$$\Rightarrow Ma_{(theory)} = Ma_{(practical)} = 0.5$$